1. $\lim_{x\to 0} \frac{\sqrt{x^2+9}-3}{x^2} = \lim_{x\to 0} \frac{(\sqrt{x^2+9}-3)(\sqrt{x^2+9}+3)}{x^2(\sqrt{x^2+9}+3)} = \lim_{x\to 0} \frac{x^2+9-9}{x^2(\sqrt{x^2+9}+3)} = \lim_{x\to 0} \frac{1}{\sqrt{x^2+9}+3} = \frac{1}{6}$

2. $\lim_{x\to 0} x \sin(\frac{1}{x}) = 0$

3. $\lim_{x\to +\infty} \frac{x}{1+x^2} = \lim_{x\to \infty} \frac{\frac{1}{x}\cdot x^2}{(\frac{1}{x^2}+1)x^2} = \lim_{x\to \infty} \frac{\frac{1}{x}}{\frac{1}{x^2}+1} = 1$

4. $\lim_{x\to 0^-} e^{1/x} = 0$ 趋近于0，不是0

(a) $\lim_{t\to 0} \frac{\sin t}{t} = 1$

5. $\lim_{x\to 0^+} x^x = 0$ (?) $0^0 = 0$

$= e^{\ln x^x} = e^{x\ln x}$, $x = \frac{1}{t}$, $\ln x = -\ln t$

$e^{\frac{1}{t}(-\ln t)} = e^{-\frac{\ln t}{t}}$ (as $x\to 0$, $t\to \infty$)

$\lim_{t\to \infty} e^{-\frac{\ln t}{t}} = e^0 = 1$

分子增长远慢于分母

6. $\frac{d}{dx}\tan x = \sec^2 x$

$\frac{d}{dx}\cot x = -\csc^2 x$

$\frac{d}{dx}\sec x = \sec x \tan x$

$\frac{d}{dx}\csc x = \csc x \cdot \cot x$

7. $\frac{d}{dx} 2^{\sec x} = (2^{\sec x}) \cdot \sec x \tan x$ $(a^x)' = a^x \ln a$

a^x $2^{\sec x} \ln 2$

8. $(\log_a x)' = \frac{1}{x \ln a}$

9. $\lim_{x\to \infty} (\frac{x-2}{x})^x = \lim_{x\to \infty} [\frac{x(1-\frac{2}{x})}{x}]^x = 1$

$\lim_{x\to 0} (1+x)^{\frac{1}{x}} = e$

$\lim_{x\to \infty} (1+\frac{1}{x})^x = e$

$= \lim_{x\to \infty} (1-\frac{2}{x})^x = \lim_{x\to \infty} [(1-\frac{2}{x})^{-\frac{x}{2}}]^{-2} = e^{-2}$